

Unit 2: Natural Resources, Energy & Conservation

Handwritten Student Notebook

Syllabus Topics Covered:

OFFICIAL SYLLABUS TOPICS COVERED Resource Foundations: Renewable vs. non-renewable resources, sustainable management Forest Resources: Uses, deforestation causes, mining effects, dams & tribal displacement, Chipko/Appiko Water Resources: Over-exploitation, groundwat...

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1. Natural Resources & Sustainability

Definition:

A **Natural Resource** is any raw material, physical matter, or energy flow provided by the Earth that is essential or useful for sustaining human civilization and ecological integrity.

Classification	Definition & Replenishment Rate	Core Examples
Renewable Resources (Inexhaustible / Regenerative)	Resources replenished naturally through biogeochemical or physical cycles at a rate equal to or faster than consumption.	Solar radiation, wind kinetic energy, tidal flow, biomass (when harvested below maximum sustainable yield), freshwater.
Non-Renewable Resources (Exhaustible)	Finite geological reserves formed over millions of years that cannot regenerate on human timescales once extracted.	Fossil fuels (coal, petroleum, natural gas), metallic ores (iron, bauxite, copper), uranium, non-metallic minerals.

Quick Recall:

Renewable = regenerated by natural cycles (solar, wind, water). Non-renewable = finite geological reserves (fossil fuels, metallic minerals).

2. Forest Resources & Deforestation

Forests are Earth's lungs, covering approximately **31%** of global land area. They serve critical economic functions (timber, pulp, resin, medicinal herbs) and invaluable ecological services (carbon sequestration, oxygen generation, hydrological regulation, soil stabilization).

2.1 Deforestation: Causes & Ecological Fallout

- **Principal Causes:** Agricultural expansion (slash-and-burn / shifting cultivation), commercial timber logging, open-cast mining in forested belts, large-scale hydroelectric reservoirs, urban infrastructure corridors.
- **Consequences:** Accelerated soil erosion, catastrophic loss of biodiversity, disruption of regional microclimate and rainfall cycles, increased atmospheric CO_2 , and displacement of indigenous forest communities.

2.2 Dams & Forest Conservation Movements

- **Dams & Tribal Displacement:** Mega-dam projects submerge thousands of hectares of prime virgin forest (e.g. Sardar Sarovar, Tehri Dam) and forcibly displace indigenous tribal populations without adequate rehabilitation.
- **The Chipko Movement (1973, Uttarakhand):** Led by Sunderlal Bahuguna and Chandi Prasad Bhatt; villagers (notably rural women) hugged trees to prevent commercial loggers from cutting them down.
- **The Appiko Movement (1983, Western Ghats, Karnataka):** Inspired by Chipko, villagers in Uttara Kannada embraced forest trees with the slogan "Ubsu, Belasu, Balasu" (Save, Grow, and Wisely Use).

Quick Recall:

Deforestation causes soil loss, climate shifts, and tribal displacement. Grassroots conservation: Chipko (Uttarakhand, Sunderlal Bahuguna) and Appiko (Karnataka Western Ghats).

3. Water Resources & Rainwater Harvesting

Freshwater constitutes only 2.5% of global water reserves. Human extraction for agriculture (70%), industry (20%), and municipal use (10%) is creating severe global water scarcity.

3.1 Over-utilization & Groundwater Depletion

- **Over-pumping:** Deep tube-wells extract groundwater faster than monsoon recharge rates, plunging water tables downward by meters per year in agricultural breadbaskets (Punjab, Karnataka).
- **Land Subsidence & Contamination:** Excessive pumping causes aquifer compaction, coastal saltwater intrusion, and mobilization of toxic geogenic fluoride and arsenic into drinking supplies.
- **Inter-State River Conflicts:** Severe geopolitical disputes over shared river basins during dry spells (e.g. the **Cauvery River Dispute** between Karnataka and Tamil Nadu, and the **Krishna Water Dispute**).

3.2 Rainwater Harvesting (RWH)

Rainwater harvesting captures and stores rainwater for direct use or for recharging subterranean aquifers via recharge pits and borewells.

Formula: Annual Rooftop Rainwater Harvest Potential:

$$V = A \times R \times C$$

where:

- V : Total harvestable water volume (m^3).
- A : Catchment roof area (m^2).
- R : Annual rainfall (m or $mm/1000$).
- C : Runoff coefficient (fraction of rainfall not absorbed; $C \approx 0.80 - 0.90$ for concrete roofs, 0.60 for tiled roofs).

Quick Recall:

Over-extraction causes aquifer depletion and river disputes (Cauvery). Rooftop harvesting yield formula: $V = A \times R \times C$ (m^3).

4. Mineral Resources & Mining Impacts

Definition:

A **Mineral** is a naturally occurring, inorganic solid substance possessing a definite chemical composition and an ordered internal crystalline atomic structure.

4.1 Environmental Fallout of Mining

- **Landscape Destruction:** Open-cast surface mining strips overburden vegetation, creating massive craters and mine dumps.
- **Acid Mine Drainage (AMD):** When sulfide minerals (e.g. pyrite, FeS_2) are exposed to air and water during mining, bacterial oxidation generates sulfuric acid (H_2SO_4), which leaches toxic heavy metals into streams:



- **Air & Noise Pollution:** Blasting, drilling, and hauling generate fugitive particulate matter ($PM_{2.5}$, PM_{10}) causing silicosis in miners.
- **Tailings Dam Failure:** Collapse of wet tailing ponds releases millions of tons of toxic slimes into river networks.

Quick Recall:

5. Food Resources & Modern Agriculture

The Green Revolution dramatically increased grain yields through High-Yield Varieties (HYVs), synthetic fertilizers, chemical pesticides, and intensive canal irrigation, but triggered severe ecological side effects:

Modern Agricultural Practice	Environmental Degradation Mechanism	Ecological / Health Impact
Excessive Synthetic Fertilizers (Nitrates & Phosphates)	Runoff into surface lakes causes Eutrophication (algal blooms, oxygen depletion, fish kills). Groundwater leaching causes Blue Baby Syndrome (Methemoglobinemia).	Aquatic dead zones; impaired oxygen delivery in infants.
Persistent Organochlorine Pesticides (DDT)	Lipid-soluble, non-biodegradable toxins accumulate up food chains via Biomagnification (concentration amplifies $\times 10^5$ from water to apex raptors).	Eggshell thinning in birds, endocrine disruption, human carcinogenicity.
Intensive Canal Flood Irrigation	Excessive water raises water table close to surface (Waterlogging). Evaporation concentrates dissolved mineral salts in topsoil (Salinization).	Osmotic stress, plant root asphyxiation, cropland desertification.

Quick Recall:

Agricultural side effects: Eutrophication & Blue Baby Syndrome (nitrates); Biomagnification (DDT concentration amplification); Waterlogging & Salinization (over-irrigation).

6. Land Resources & Soil Erosion

Soil is a non-renewable resource on human timescales, requiring centuries to form a single centimeter of topsoil through bedrock weathering.

6.1 Mechanisms & Types of Soil Erosion

A. Wind Erosion (Exam Focus: CIE 1 2024 Question):

- **1. Saltation (50 – 70% of wind erosion):** Fine-to-medium sand particles (0.1 – 0.5 mm) bounce along the surface in short hops, dislodging other particles upon impact.

- **2. Surface Creep (5 – 25% of wind erosion):** Larger particles (0.5 – 2.0 mm) too heavy to be lifted are rolled or pushed along the ground by colliding saltating grains.
- **3. Suspension (3 – 15% of wind erosion):** Extremely fine silt and dust particles (< 0.1 mm) are lifted high into the atmosphere and transported thousands of kilometers as dust storms.

B. Water Erosion:

- **Sheet Erosion:** Uniform detachment and thin horizontal removal of topsoil by overland sheet runoff.
- **Rill Erosion:** Concentrated runoff cuts small, shallow, distinct finger-like channels (rills).
- **Gully Erosion:** Rills converge into deep, steep-sided channels (gullies) that dissect croplands into uncultivable ravines (badland topography).

6.2 Soil Conservation Measures

- **Contour Bunding & Terracing:** Sculpting steep slopes into stepped terraces to slow surface runoff velocity.
- **Afforestation & Shelterbelts:** Planting dense tree windbreaks perpendicular to prevailing winds to block wind erosion.
- **Mulching & Cover Cropping:** Retaining crop residues on soil to prevent direct raindrop impact detachment.

Quick Recall:

Wind erosion types: Saltation (bouncing grains, 50–70%), Surface creep (rolling coarse grains), Suspension (fine dust airborne). Water erosion: Sheet → Rill → Gully.

7. Energy Resources (Fossil Fuels vs. Renewable Alternatives)

7.1 Environmental Fallout of Fossil Fuels

- **Coal Combustion:** Emits large amounts of CO_2 (greenhouse gas driving global warming), SO_2 and NO_x (causing **Acid Rain**, $pH < 5.6$), particulate matter, and toxic fly ash containing heavy metals (mercury, lead).
- **Petroleum / Transportation:** Major source of photochemical smog (ground-level ozone, **PAN**), carbon monoxide (CO), and marine oil spills.

- **Nuclear Power:** Zero greenhouse gas emissions during operation, but carries catastrophic risk of core meltdown (Chernobyl 1986, Fukushima 2011) and unsolved challenges of long-term deep geological disposal of high-level radioactive waste (plutonium isotopes with half-lives exceeding 24,000 years).

7.2 Renewable Energy Alternatives

- **Solar Energy:** Photovoltaic (PV) cells convert sunlight directly into electricity via silicon semiconductor p-n junctions; solar thermal collectors generate heat/steam.
- **Wind Energy:** Aerodynamic wind turbine blades capture atmospheric kinetic energy; clean, modular, zero operational water use.
- **Biomass & Biogas:** Anaerobic digestion of animal manure and organic waste by methanogenic bacteria produces **Biogas** (55 – 70% CH_4 , 30 – 45% CO_2), providing clean domestic cooking fuel while yielding nitrogen-rich organic slurry fertilizer.
- **Hydroelectric & Geothermal:** Hydro harness gravitational potential of flowing water; Geothermal taps subterranean hydrothermal steam.

Quick Recall:

Fossil fuels drive acid rain (SO_2 , NO_x) and global warming (CO_2). Renewables (Solar PV, Wind, Biogas [methane]) provide decentralized, low-carbon power.

8. Solved Environmental Calculations

Solved Problem 2.1: Rooftop Rainwater Harvesting Yield:

Problem Statement: A residential campus building has a flat concrete rooftop of dimensions $15 \text{ m} \times 10 \text{ m}$. The annual rainfall in the region is 900 mm . If the runoff coefficient for the concrete roof is $C = 0.85$, calculate:

1. The total volume of rainwater that can be harvested annually in cubic meters (m^3).
2. The total volume in liters.

Step 1: Compute Catchment Roof Area (A):

$$A = 15 \text{ m} \times 10 \text{ m} = 150 \text{ m}^2$$

Step 2: Convert Rainfall to Meters (R):

$$R = \frac{900 \text{ mm}}{1000 \text{ mm/m}} = 0.90 \text{ m}$$

Step 3: Apply the Harvesting Formula ($V = A \times R \times C$):

$$V = 150 \text{ m}^2 \times 0.90 \text{ m} \times 0.85 = 135 \times 0.85 = 114.75 \text{ m}^3$$

Step 4: Convert to Liters ($1 \text{ m}^3 = 1,000 \text{ Liters}$):

$$V_{\text{litres}} = 114.75 \times 1,000 = 114,750 \text{ Liters}$$

$$V = 114.75 \text{ m}^3 = 114,750 \text{ Liters / year}$$

Impact: This harvested volume can fulfill the 135 L/capita/day domestic drinking and sanitary needs of a 4-person family for over 212 days!

Solved Problem 2.2: Solar Rooftop PV Generation and Annual Carbon Offset:

Problem Statement: An institution installs a 5.0 kWp rooftop Solar PV system. In Karnataka, each 1 kWp generates an average of 4.5 kWh of electricity per day. If the Indian coal-dominated power grid has an average emission factor of 0.82 kg CO₂/kWh, calculate:

1. The total electrical energy generated by the system annually.
2. The annual carbon dioxide (CO₂) emission offset in metric tons.

Step 1: Annual Electricity Generation:

$$E_{\text{annual}} = 5.0 \text{ kWp} \times 4.5 \text{ kWh/kWp/day} \times 365 \text{ days} = 8,212.5 \text{ kWh}$$

Step 2: Annual Carbon Dioxide Offset:

$$\text{Offset} = 8,212.5 \text{ kWh} \times 0.82 \text{ kg CO}_2/\text{kWh} = 6,734.25 \text{ kg CO}_2$$

Step 3: Convert to Metric Tons (1 metric ton = 1,000 kg):

$$\text{Offset}_{\text{tons}} = \frac{6,734.25}{1,000} \approx 6.734 \text{ metric tons of CO}_2$$

$$E_{\text{annual}} = 8,212.5 \text{ kWh}, \quad \text{Annual CO}_2 \text{ Offset} = 6.734 \text{ Metric Tons}$$

INTERACTIVE LAB EXPLORER

Water Harvesting & Solar Carbon Offset Studio

Tab 1: Rainwater Harvesting

Tab 2: Solar PV & Carbon Offset

Calculate annual rooftop rainwater harvesting potential using the hydrological runoff equation:

$$V = A \times R \times C$$

Roof Catchment Area (A):

150 m²

Annual Rainfall (R):

900 mm/yr (0.90 m)

Roof Surface / Runoff Coeff (C):

Reinforced Concrete Roof ($C = 0.85$ - Solved Example)



HARVESTED WATER YIELD

114,750 Litres / year

Volume: 114.75 m³ (150 × 0.90 × 0.85)

Domestic Water Security Index:

Supplies a 4-person household (at 135 L/capita/day = 540 L/day) for **212.5 days** of clean domestic use!

9. Quick Revision Sheet

Resource Topic	Degradation Mechanism	Mitigation & Conservation Strategy
Forests	Deforestation, mining, mega-dams submergence.	Afforestation, Chipko Movement, Appiko Movement.
Water	Over-pumping, water table collapse, river disputes (Cauvery).	Rainwater Harvesting ($V = A \times R \times C$), recharge pits, drip irrigation.
Minerals	Acid Mine Drainage (H_2SO_4 generation), tailing dam failures.	Mine reclamation, lime neutralization of AMD, mineral recycling.
Food & Soil	Eutrophication, Biomagnification (DDT), Salinization, Erosion (Saltation).	Organic farming, contour terracing, shelterbelts, mulching.
Energy	Fossil fuel acid rain (SO_2 , NO_x), nuclear radioactive wastes.	Solar PV, Wind, Biogas (methanogenic anaerobic digestion).

10. Authentic Exam Archive & Model Solutions

The following authentic questions have been transcribed directly from official department examination papers in notes/evs/practice/evs-cie1-qp-2024.docx (Ramaiah Continuous Internal Evaluation - I, Course Code **HS 510: Environmental Studies**):

Part A: 1-Mark Objective Questions (CIE-1 2024-2025)

Practice Problem:

CIE-1 Q4 [1M] These are some of the adverse effects of mining:

Solution:

Official Options: (a) Water pollution · (b) Air pollution · (c) Noise pollution · (d) All of the above

Correct Answer: (d) All of the above.

Academic Explanation: Mining activities induce multiple cross-media ecological damage:

- **Water Pollution:** Acid Mine Drainage (oxidation of iron pyrite FeS_2), leaching of toxic heavy metals (Pb , Cd , As), and siltation of downstream rivers from tailings pond breaches.
- **Air Pollution:** Fugitive particulate matter emissions (PM_{10} and $PM_{2.5}$) generated by blasting, haul roads, and mineral processing crushers.
- **Noise Pollution:** High-decibel emissions and ground vibrations caused by explosive blasting, heavy earth-moving equipment, and slurry transport pumps.

Practice Problem:

CIE-1 Q5 [1M] Saltation, surface creep, and suspension are the types of soil erosion by:

Solution:

Official Options: (a) Wind · (b) Water · (c) Mining · (d) None of the above

Correct Answer: (a) Wind.

Physical Transport Mechanics:

- **Saltation (50–70% of wind transport):** Medium sand particles (0.1 – 0.5 mm) bounce in short hops along the ground.
- **Surface Creep (5–25% of transport):** Coarser soil grains (0.5 – 2.0 mm) are pushed or rolled along the surface by colliding saltating particles.
- **Suspension (3–40% of transport):** Very fine silt and clay particles (< 0.1 mm) are lifted into turbulent upper atmospheric air currents and transported over thousands of kilometers.

Practice Problem:

CIE-1 Q7 [1M] Extensive planting of trees to increase forest cover is called:

Solution:

Official Options: (a) Afforestation · (b) Agro forestry · (c) Deforestation · (d) Social forestry

Correct Answer: (a) Afforestation.

Explanation: Afforestation is the establishment of a forest or stand of trees in an area where there was no recent previous tree cover, serving as a carbon sink and halting desertification.

Practice Problem:

CIE-1 Q8 [1M] Sustainability requires:

Solution:

Official Options: (a) Conservation of resources · (b) Minimizing depletion of non-renewable resources · (c) Using sustainable practices for managing renewable resources · (d) All

Correct Answer: (d) All.

Explanation: Ecological sustainability requires balancing consumption with regeneration rates so present societal needs are met without compromising future generations.

Practice Problem:

CIE-1 Q9 [1M] Which of the following is the most environmentally friendly agriculture practice?

Solution:

Official Options: (a) Using chemical fertilizers · (b) Using insecticides · (c) Organic farming · (d) None of the above

Correct Answer: (c) Organic farming.

Explanation: Organic farming eliminates synthetic agrochemicals, relying on biofertilizers (compost, vermicompost, Rhizobium), crop rotation, and biological pest control to preserve soil microbial biodiversity and prevent groundwater nitrate toxicity.

Practice Problem:

CIE-1 Q10 [1M] A mineral is defined as a:

Solution:

Official Options: (a) Organic matter · (b) Naturally occurring inorganic substance · (c) Synthetic compound · (d) None

Correct Answer: (b) Naturally occurring inorganic substance.

Explanation: A mineral is a naturally occurring, homogeneous, inorganic solid having a definite chemical composition and an ordered internal crystalline structure.

Part B: 5-Mark Subjective Questions (CIE-1 2024-2025)Practice Problem:

CIE-1 Q1.b [5M] What are the environmental effects of using fossil fuels?

Solution:

Official Model Solution: The combustion of fossil fuels (coal, petroleum, natural gas) generates severe multidimensional impacts:

1. **Anthropogenic Greenhouse Effect & Global Warming:** Burning carbon yields massive CO_2 emissions (rising from pre-industrial 280 ppm to over 420 ppm), driving global temperature rise, polar ice melt, and sea-level rise.
2. **Acid Deposition (Acid Rain):** High sulfur and nitrogen content in coal/petroleum combusts into SO_2 and NO_x , which oxidize into sulfuric acid (H_2SO_4) and nitric acid (HNO_3), dropping precipitation $pH < 5.6$ and corroding monuments and acidifying freshwater fisheries.
3. **Photochemical Smog & Respiratory Distress:** Vehicle exhausts release volatile organic compounds (VOCs) and NO_x , reacting with sunlight to form ground-level ozone (O_3) and photochemical smog, triggering asthma and chronic bronchitis.
4. **Thermal Pollution & Ash Disposal:** Coal-fired thermal power stations produce millions of tons of fly ash containing toxic heavy metals (Pb , As , Hg), contaminating groundwater aquifers and topsoils.

Practice Problem:

CIE-1 Q2.b [5M] Explain the effects of overuse of fresh water and problems related to water resources.

Solution:

Official Model Solution: Over-exploitation of freshwater leads to critical hydrological crises:

1. **Water Table Depletion:** Intensive tube-well extraction for green-revolution agriculture exceeds natural monsoon recharge, causing regional aquifer water tables to fall by meters annually.
2. **Saltwater Intrusion:** Excessive drawdown in coastal areas reverses hydraulic gradients, pulling dense saline seawater into freshwater municipal aquifers and ruining drinking water supplies.
3. **Land Subsidence:** Over-extraction drains interstitial pore water from deep silt/clay strata, causing the overburden ground to compact and sink irreversibly (e.g., Mexico City, Jakarta, Venice).
4. **Waterlogging and Soil Salinization:** In canal-irrigated zones, excessive irrigation without adequate drainage raises the water table into the root zone, causing capillary action to evaporate water and leave a toxic crust of alkaline salts on arable soils.
5. **Riparian & Interstate Water Conflicts:** Upstream impoundments restrict downstream flows during dry seasons, causing severe socio-economic disputes (e.g. Cauvery River dispute between Karnataka and Tamil Nadu).

Dimension	Conventional / Non-Renewable	Non-Conventional / Renewable
Sources	Coal, Crude Oil, Natural Gas, Nuclear Fission (U^{235}).	Solar PV, Wind Turbines, Small Hydro, Biomass, Geothermal.
Depletion Profile	Finite geological deposits; exhausting within decades to centuries.	Naturally replenished continuously via solar and planetary cycles.
Carbon Footprint	High greenhouse gas and particulate emissions (0.82 – 1.0 kg CO ₂ /kWh).	Zero or near-zero operational greenhouse gas emissions.

Dimension	Conventional / Non-Renewable	Non-Conventional / Renewable
Ecological Hazards	Acid rain, oil spills, radioactive nuclear waste storage.	Intermittent output, land requirement, bird strike risks (wind).

Verified Numerical & Conceptual Practice Problems

PRACTICE 1 · ROOFTOP RAINWATER HARVESTING YIELD

Calculate the annual harvested rainwater yield for a concrete rooftop of area 150 m^2 located in a region with 900 mm annual rainfall and runoff coefficient 0.85 .

Practice Problem:

Show Step-by-Step Solution

Solution:

Applying the hydrological harvesting equation: $V = A \times R \times C$

- Catchment Area: $A = 150 \text{ m}^2$
- Rainfall: $R = 900 \text{ mm} = 0.90 \text{ m}$
- Runoff Coefficient: $C = 0.85$ (concrete roof)

$$V = 150 \text{ m}^2 \times 0.90 \text{ m} \times 0.85 = 135 \times 0.85 = 114.75 \text{ m}^3$$

$$V_{\text{litres}} = 114.75 \times 1,000 = 114,750 \text{ Litres}$$

Verification: Verified by automated unit test in
audit/verify/evs/verify_evs_u2.py.

PRACTICE 2 · SOLAR PV GENERATION & CO₂ OFFSET

A 5.0 kW_p solar PV rooftop system operates with an average yield of 4.5 kWh/kW_p/day. If the grid emission factor is 0.82 kg CO₂/kWh, calculate annual energy generated and metric tons of CO₂ offset.

Practice Problem:

Show Step-by-Step Solution

Solution:**1. Annual Electricity Generation:**

$$E_{\text{annual}} = 5.0 \text{ kW}_p \times 4.5 \text{ kWh/kW}_p/\text{day} \times 365 = \mathbf{8,212.5}$$

2. Annual Carbon Dioxide Offset:

$$\text{CO}_2 \text{ Offset} = 8,212.5 \text{ kWh} \times 0.82 \text{ kg CO}_2/\text{kWh} = \mathbf{6,734.25}$$

3. Conversion to Metric Tons:

$$\text{Offset}_{\text{tons}} = \frac{6,734.25}{1,000} = \mathbf{6.734} \text{ Metric Tons}$$

Verification: Verified by automated unit test in
audit/verify/evs/verify_evs_u2.py.

PRACTICE 3 · ACID MINE DRAINAGE (AMD) REACTION

Write the stoichiometric chemical reaction for Acid Mine Drainage (AMD) formation and explain why it severely pollutes aquatic ecosystems.

Practice Problem:

Show Step-by-Step Solution

Solution:

When sub-surface mining exposes iron pyrite (FeS_2) to atmospheric oxygen and groundwater, iron-oxidizing bacteria (Acidithiobacillus ferrooxidans) catalyze the reaction:



Ecological Impact: Generates concentrated sulfuric acid (H_2SO_4), driving stream pH down to 2 — 3, dissolving highly toxic heavy metals (aluminum, lead, cadmium), and suffocating aquatic gills with insoluble orange-brown ferric hydroxide precipitations ("yellow boy").

PRACTICE 4 · WATERLOGGING VS SALINIZATION MECHANISMS

Differentiate between waterlogging and soil salinization in agricultural land management.

Practice Problem:

Show Step-by-Step Solution

Solution:

- **Waterlogging:** Physical condition where excessive irrigation water saturates soil pore spaces, pushing the water table into the plant root zone, displacing oxygen, and causing anaerobic root asphyxiation.
- **Soil Salinization:** Chemical degradation where high water tables evaporate in arid/semi-arid climates, drawing dissolved minerals upward via capillary action and depositing a sterile white crust of alkaline sodium, calcium, and magnesium salts on topsoil.

Mode of Erosion	Particle Diameter Range	Percentage of Total Wind Transport
Saltation (Bouncing)	0.1 mm to 0.5 mm	50% to 70% (Dominant mechanism)
Surface Creep (Rolling)	0.5 mm to 2.0 mm	5% to 25%
Suspension (Airborne)	< 0.1 mm (Fine silt/clay)	3% to 40%

PRACTICE 6 · ECOLOGICAL CONFLICTS OF MEGA DAMS

Analyze the key environmental and social trade-offs of large river valley projects with reference to Sardar Sarovar and Tehri Dams.

Practice Problem:

Show Step-by-Step Solution

Solution:

While multipurpose mega dams generate hydroelectricity and irrigation water, they incur severe ecological and socio-economic trade-offs:

- **Forest & Biodiversity Submergence:** Inundation of vast tracts of pristine riverine forests, destroying wildlife corridors and endemic habitats.
- **Tribal Displacement & Resettlement Trauma:** Uprooting indigenous tribal communities from ancestral lands without adequate rehabilitation (the core focus of Medha Patkar's Narmada Bachao Andolan).
- **Reservoir-Induced Seismicity (RIS):** Massive concentrated water masses in seismic zones (e.g. Tehri Dam in the Central Himalayan seismic gap) create pore pressure lubrication along active fault lines.
- **Silt Trap & Downstream Delta Erosion:** Trapping fertile nutrient silts in reservoirs deprives downstream estuaries, starving agricultural deltas and triggering coastal erosion.

11. Prescribed References

Course Reference Literature:

1. **Erach Bharucha**, Textbook of Environmental Studies for Undergraduate Courses, Universities Press / UGC India. (Chapter 2: Natural Resources).
2. **Benny Joseph**, Environmental Studies, Tata McGraw-Hill Education.
3. **R. Rajagopalan**, Environmental Studies: From Crisis to Cure, Oxford University Press.

AI-assisted study notes written from the official courseware. Verify against your faculty's presentation slides before the exam.